

Climate and Migration in the United States

Structural Adaptation and Spatial Equilibrium Realignment

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Core Analytical Problem:

- Existing literature predominantly evaluates fast-onset climate disasters (hurricanes, floods) or agricultural shocks in developing economies.
- This paper investigates **slow-onset climate change** (persistent temperature shifts) within a high-income, climate-shielded institutional context (the United States).

Spatial Equilibrium Mechanism (Roback, 1982):

- Household location maximizes utility subject to constraints of wages, rents, and local amenities.
- *Hypothesis*: In advanced economies, extreme heat operates less as a severe income shock (due to a non-agricultural baseline and air conditioning) and more as a deterioration of **amenity value**, prompting structural spatial reallocation.

Data Architecture: Dual-Frequency Measurement

To isolate causal responses across time scales, the authors construct two distinct datasets:

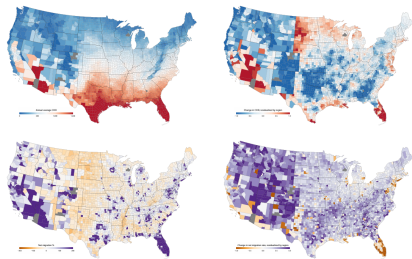
1. Decadal Migration (Census, 1950-2010):

- Derived via the “forward residual method”:
$$\text{Population}_t - (\text{Population}_{t-1} + \text{Births}_t - \text{Deaths}_t).$$
- *Advantage*: High precision, near-universal coverage. Eliminates attenuation bias.
- *Constraint*: Net migration only; cannot separate inflows from outflows. Medium-term frequency.

2. Annual Migration (IRS, 1983-2018):

- Based on year-to-year address changes on tax filings.
- *Advantage*: Decomposes net migration into distinct in-migration (pull) and out-migration (push) flows. High-frequency (annual).
- *Constraint*: Underrepresents non-filers (low-income, elderly).

Spatial Distribution and Cross-Sectional Confounders



Identifying the Variation:

- *Left Panels:* Cross-sectional averages. Shows historic “hollowing-out” of the US interior and migration to warmer coasts/South. This is heavily confounded by unobservable economic trends.
- *Right Panels:* Within-county changes (1950s to 2000s) residualized by Census Region.
- *Causal Mandate:* Must strip away the cross-sectional “heat-seeking” trend to isolate the penalty of **locally unusual** temperature increases.

Causal Identification 1: The Panel Model

To isolate short-to-medium term structural responses, the authors estimate a fixed-effects panel model:

$$\text{Migration rate}_{it} = \beta_H \text{HDD}_{it} + \beta_C \text{CDD}_{it} + \beta_P \text{Precip}_{it} + \phi_i + \phi_{rt} + \epsilon_{it} \quad (1)$$

Design Rationale:

- $\text{HDD}_{it}, \text{CDD}_{it}$: Heating/Cooling Degree Days measure exposure to extreme temperatures relative to a 18.3°C baseline.
- ϕ_i (County FE): Absorbs time-invariant local characteristics (baseline amenities, geography).
- ϕ_{rt} (Region-by-period FE): Absorbs region-wide macroeconomic shocks or secular migration trends.
- *Assumption*: Residual temperature variation within a county-period is orthogonal to unobserved local economic shocks.

Causal Identification 2: Long Differences (LD) Model

To capture permanent adaptation and filter out short-term adjustment frictions, the authors estimate long-run changes:

$$\overline{\Delta \text{Migration rate}}_i = \beta_H \overline{\Delta \text{HDD}}_i + \beta_C \overline{\Delta \text{CDD}}_i + \beta_P \overline{\Delta \text{Precip}}_i + \phi_r + \epsilon_i \quad (2)$$

Design Rationale:

- Compares average conditions at the start of the sample (e.g., 1950s) to the end (e.g., 2000s).
- Identifies the parameter from low-frequency, persistent shifts in local climate.
- Solves the delayed response problem: household relocation carries high fixed costs; short-term weather anomalies may not cross the threshold to trigger migration, but permanent climate shifts will.

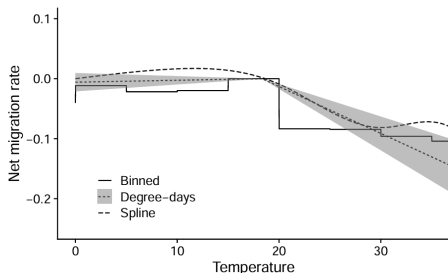
Decadal Estimates (Census): The Magnitude of Adaptation

	Net migration rate (%)			
	Panel		LD	
	(1)	(2)	(3)	(4)
HDD (100s)	-0.02 (0.04)	-0.03 (0.04)	0.19 (0.11)	-0.11 (0.13)
CDD (100s)	-0.91 (0.11)	-0.78 (0.12)	-1.20 (0.20)	-1.57 (0.24)
Precip. (100s mm)	-0.08 (0.03)	-0.09 (0.04)	-0.04 (0.09)	0.00 (0.09)
<i>Fixed effects</i>				
County	✓	✓		
Decade	✓			
Region-Decade		✓		
Region				✓
Observations	18,306	18,306	3,051	3,051

Interpretation (Table 1):

- **Panel (Col 2):** +100 CDDs decreases annualized net migration by 0.78 percentage points (≈ 0.5 SD).
- **LD (Col 4):** +100 CDDs decreases net migration by 1.57 percentage points (≈ 0.92 SD).
- *Mechanism check:* The response to temperature increases significantly as the time horizon expands. Households require time to update expectations from "transitory weather" to "permanent climate state."

Robustness: Functional Form of Temperature



Non-linear Penalty (Figure 2):

- Replaces linear CDD/HDD with 5-degree bins or a cubic spline.
- Results confirm that the migration penalty accrues specifically at the upper tail of the temperature distribution ($> 20^{\circ}\text{C}$ daily average).
- Validates the use of the degree-day (CDD) formulation as a precise linear proxy for extreme heat exposure.

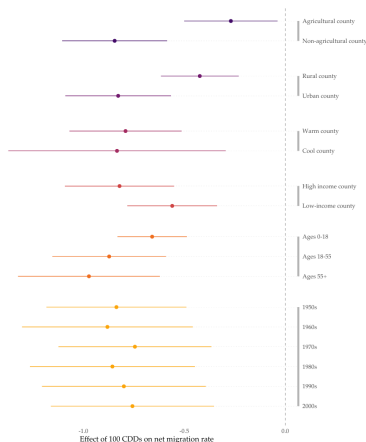
Annual Estimates (IRS): Decomposing the Flows

	Net migration rate (%)		Out-migration rate (%)		In-migration rate (%)	
	Panel	LD	Panel	LD	Panel	LD
	(1)	(2)	(3)	(4)	(5)	(6)
HDD (100s)	-0.01 (0.01)	-0.14 (0.05)	0.02 (0.01)	0.19 (0.09)	0.01 (0.01)	0.04 (0.09)
CDD (100s)	-0.02 (0.01)	-0.40 (0.07)	-0.01 (0.01)	0.20 (0.09)	-0.04 (0.01)	-0.20 (0.10)
Precip. (100 mm)	-0.01 (0.00)	0.02 (0.03)	0.01 (0.00)	0.03 (0.04)	0.00 (0.00)	0.05 (0.04)
<i>Fixed effects</i>						
County	✓		✓		✓	
Region-Year	✓		✓		✓	
Region		✓		✓		✓
Observations	108,742	3,033	108,742	3,033	108,742	3,033

Push vs. Pull (Table 2):

- IRS data confirms the LD > Panel magnitude relationship (Col 2 vs Col 1).
- **Out-migration (Col 4):** Heat increases the "push" effect (+0.20).
- **In-migration (Col 6):** Heat decreases the "pull" effect (-0.20).
- **Conclusion:** The net population penalty is driven symmetrically. Rising temperatures repel prospective new residents just as strongly as they evict existing ones.

Mechanism Heterogeneity: Falsifying Economic Intuition



Amenity vs. Income (Figure 3):

- *Anti-intuition:* Agricultural counties do **not** drive the result. Non-agricultural and urban counties show equal or stronger negative responses to heat.
- *Age Dynamics:* The 55+ cohort exhibits the highest elasticity. Since they are largely detached from the local labor market, their migration is driven by consumption/amenities, not wage differentials.
- *Income constraint:* High-income counties react more strongly, confirming that climate migration requires liquid capital to overcome friction costs.

1. The True Cost of Adaptation:

- Relocation is a profoundly costly form of climate adaptation. Identifying significant internal migration in the US implies that current integrated assessment models (IAMs) may severely underprice the systemic costs of global warming.

2. The Future Global Pathway:

- As global GDP per capita converges upward (e.g., SSP2 projections), developing nations will shift from agriculture-dependent income shocks to amenity-driven residential sorting.
- The U.S. baseline indicates that as wealth increases, the spatial equilibrium becomes **more** sensitive to climate amenities, not less. We should anticipate massive, capital-intensive spatial reallocations globally as slow-onset warming persists.